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COURSE:DSA0602-Data handling and visualization

1;Perform Data Handling and Visualization on the Employee dataset using R. Analyze the dataset by handling data and generating different charts such as Histogram, Bar Chart, Pie Chart, Line Chart, Scatter Plot, Box Plot, Violin Plot, Density Plot, Area Chart, Correlation Matrix, and Pair Plot to understand employee characteristics and trends.

Procedure

Open Google Colab (R Runtime) or RStudio.

Install and load the required libraries (ggplot2, dplyr, corrplot, and GGally).

Import the Employee.csv dataset.

Display the structure and summary of the dataset.

Check for missing values and duplicate records.

Convert categorical variables into factor data types if necessary.

Create the following visualizations:

Histogram

Bar Charts

Pie Chart

Line Chart

Area Chart

Scatter Plot

Box Plot

Violin Plot

Density Plot

Frequency Polygon

Dot Plot

Q-Q Plot

Stacked Bar Chart

Grouped Bar Chart

Correlation Matrix

Pair Plot

Analyze the generated graphs to understand employee distribution, demographics, experience, payment tiers, and leave status.

Interpret the observations obtained from the visualizations.

Education	JoiningYear	City	PaymentTier	Age	Gender	EverBenched	ExperienceInYears	LeaveOrNot
Bachelors	2017	Bangalore	3	34	Male	No	0	0
Bachelors	2013	Pune	1	28	Female	No	3	1
Bachelors	2014	New Delhi	3	38	Female	No	2	0
Masters	2016	Bangalore	3	27	Male	No	5	1
Masters	2017	Pune	3	24	Male	Yes	2	1
Bachelors	2016	Bangalore	3	22	Male	No	0	0
Bachelors	2015	New Delhi	3	38	Male	No	0	0
Bachelors	2016	Bangalore	3	34	Female	No	2	1
Bachelors	2016	Pune	3	23	Male	No	1	0
Masters	2017	New Delhi	2	37	Male	No	2	0

```
# Load Libraries
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
# Read Dataset
```

```
employee <- read.csv(file.choose())
```

```
# Data Handling
```

```
str(employee)
```

```
summary(employee)
```

```
colSums(is.na(employee))
```

```
employee <- distinct(employee)
```

```
# Convert to Factors
```

```
employee$Education <- as.factor(employee$Education)
```

```
employee$Gender <- as.factor(employee$Gender)
employee$City <- as.factor(employee$City)
employee$LeaveOrNot <- as.factor(employee$LeaveOrNot)
```

```
# Histogram
```

```
ggplot(employee,aes(Age))+
geom_histogram(binwidth=2,fill="skyblue",color="black")+
ggtitle("Age Distribution")
```

```
# Bar Chart
```

```
ggplot(employee,aes(Education))+
geom_bar(fill="steelblue")+
ggtitle("Education")
```

```
# Pie Chart
```

```
leave <- employee %>% count(LeaveOrNot)
ggplot(leave,aes(x="",y=n,fill=LeaveOrNot))+
geom_col(width=1)+
coord_polar("y")+
ggtitle("Leave Status")
```

```
# Scatter Plot
```

```
ggplot(employee,aes(Age,ExperienceInCurrentDomain,color=LeaveOrNot))+
geom_point(size=3)+
ggtitle("Age vs Experience")
```

```
# Box Plot
```

```
ggplot(employee,aes(LeaveOrNot,Age,fill=LeaveOrNot))+
geom_boxplot()+
```

```
ggtitle("Age by Leave Status")
```

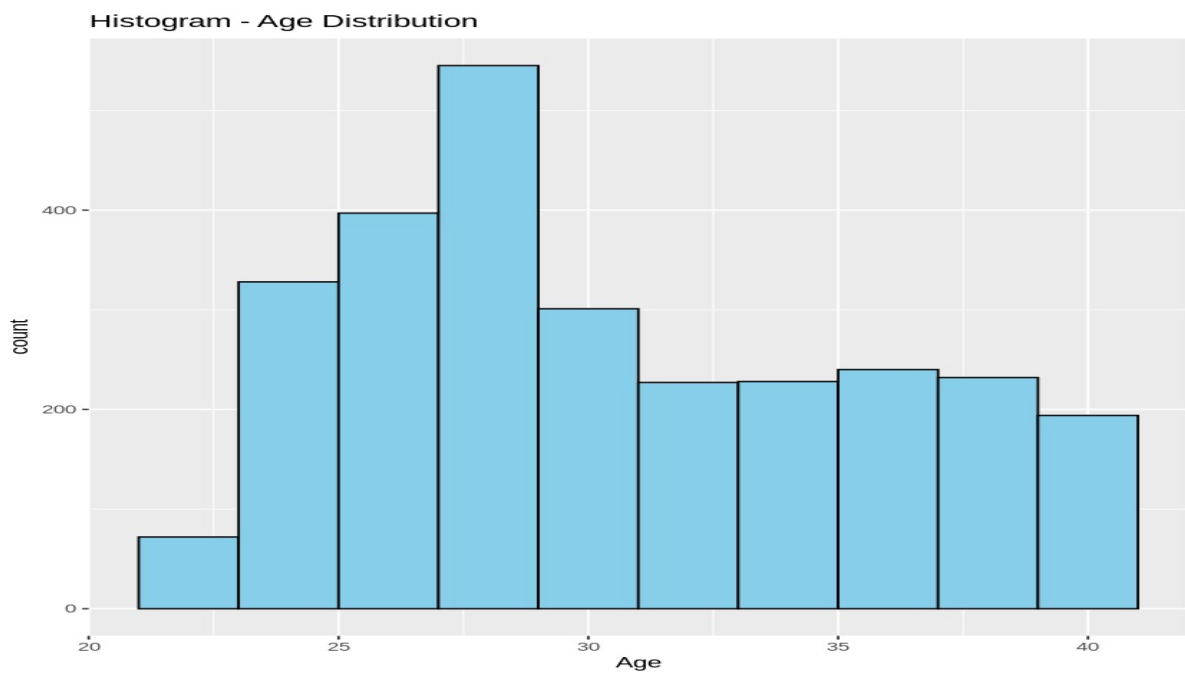
```
# Summary
```

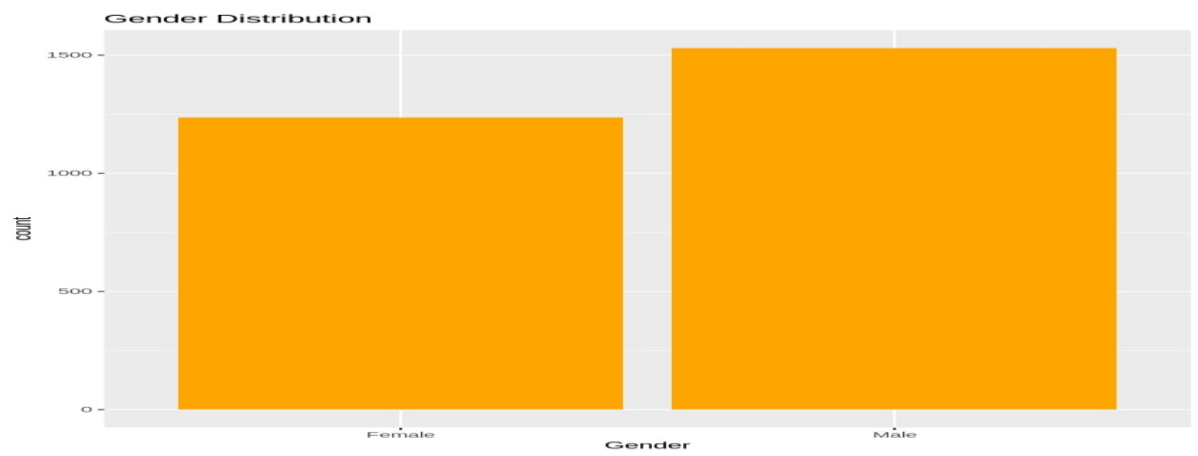
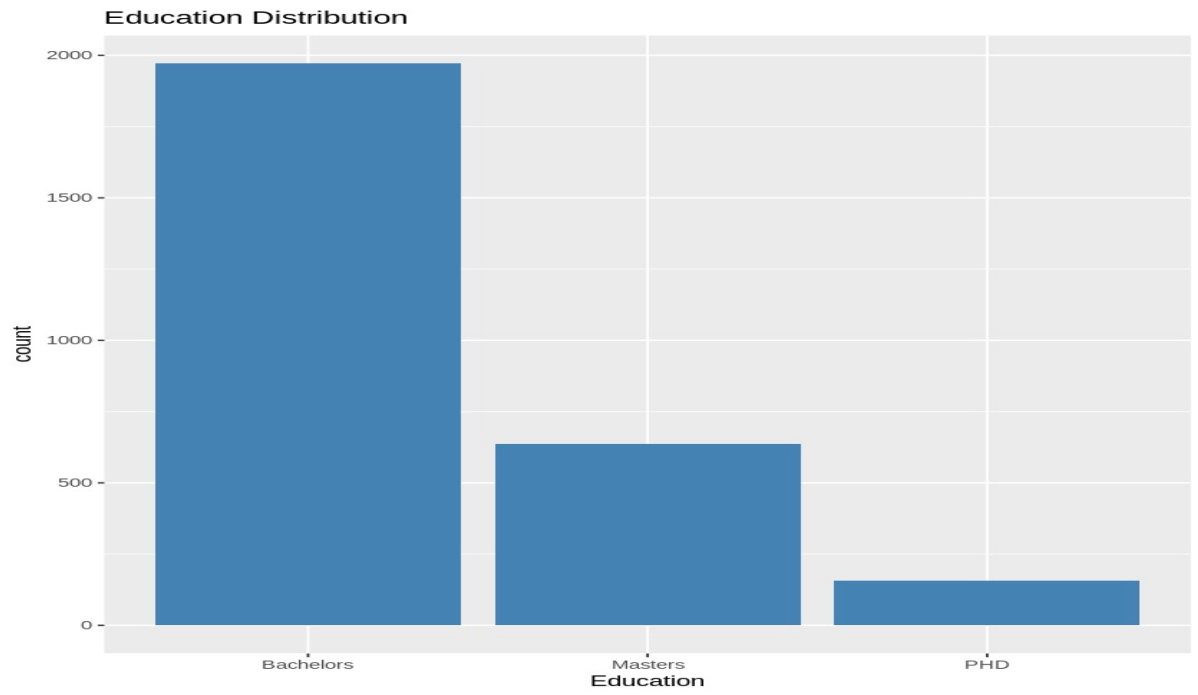
```
table(employee$Education)
```

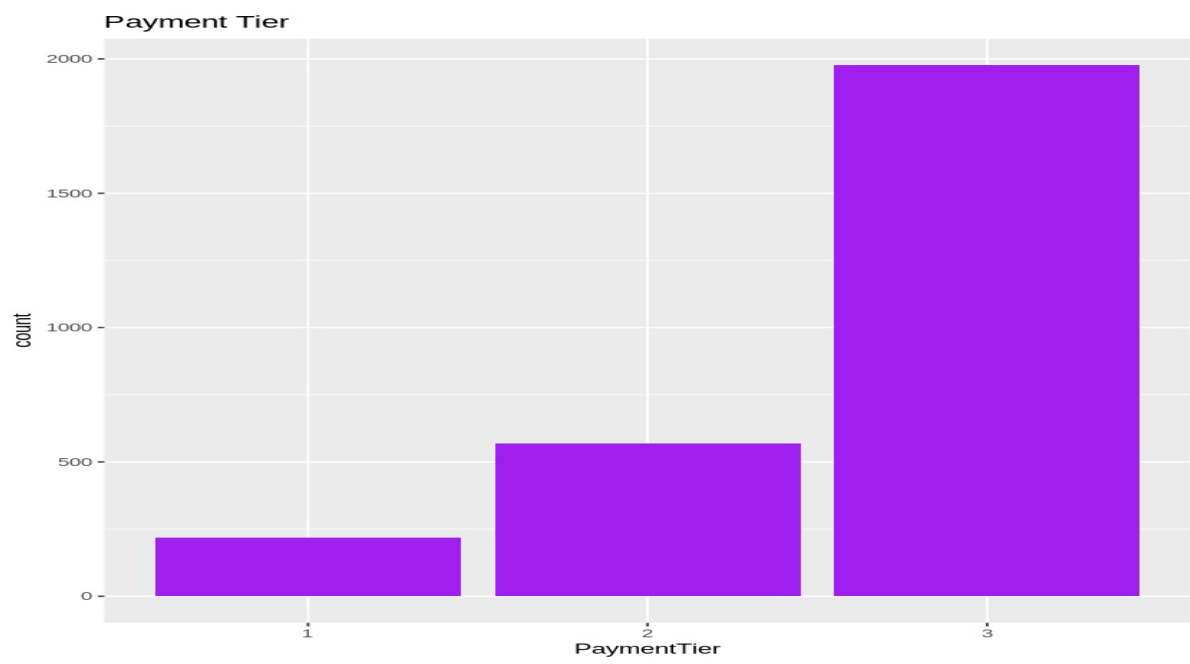
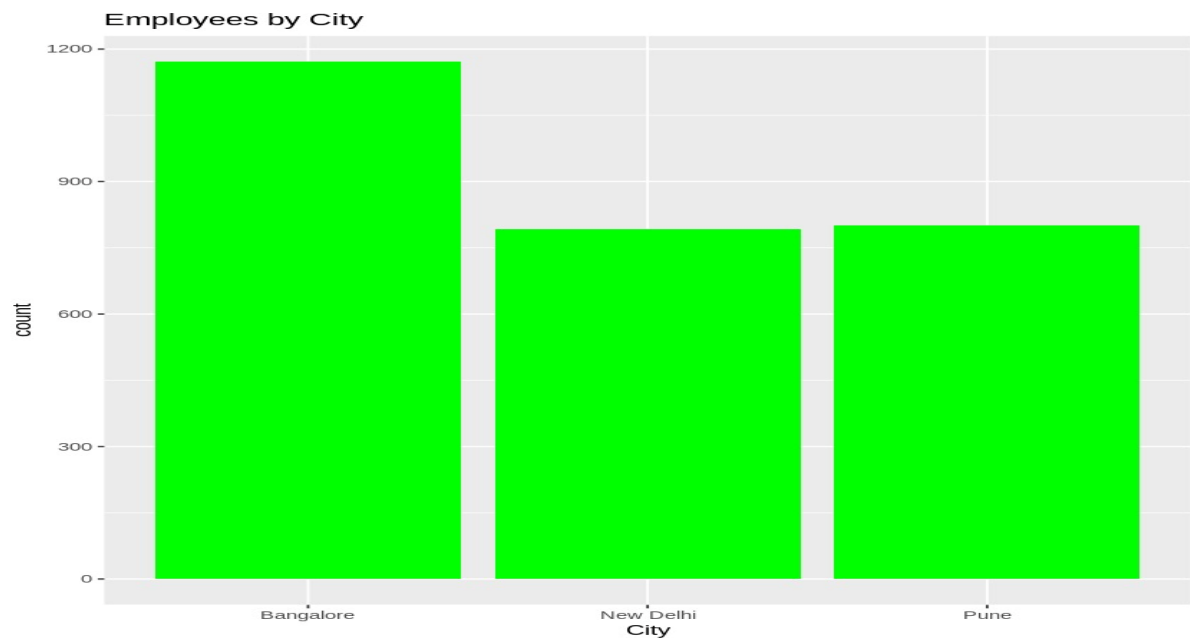
```
table(employee$Gender)
```

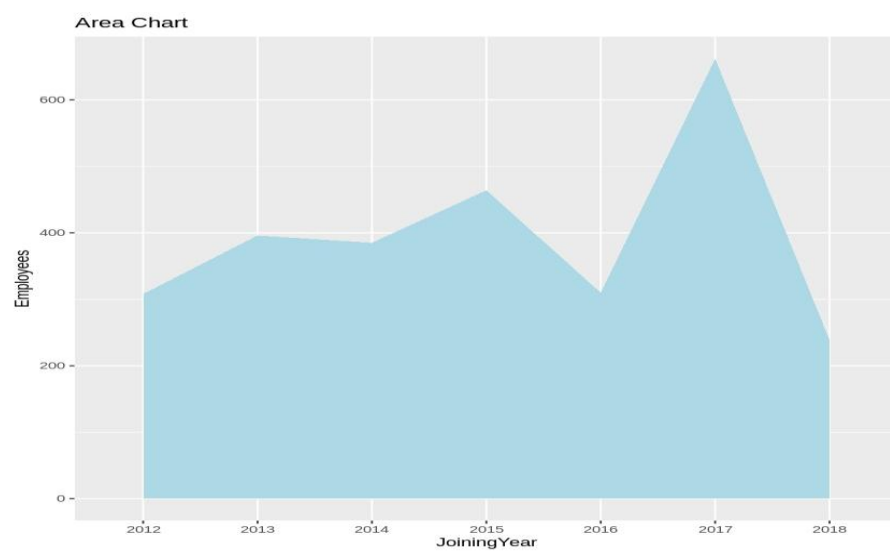
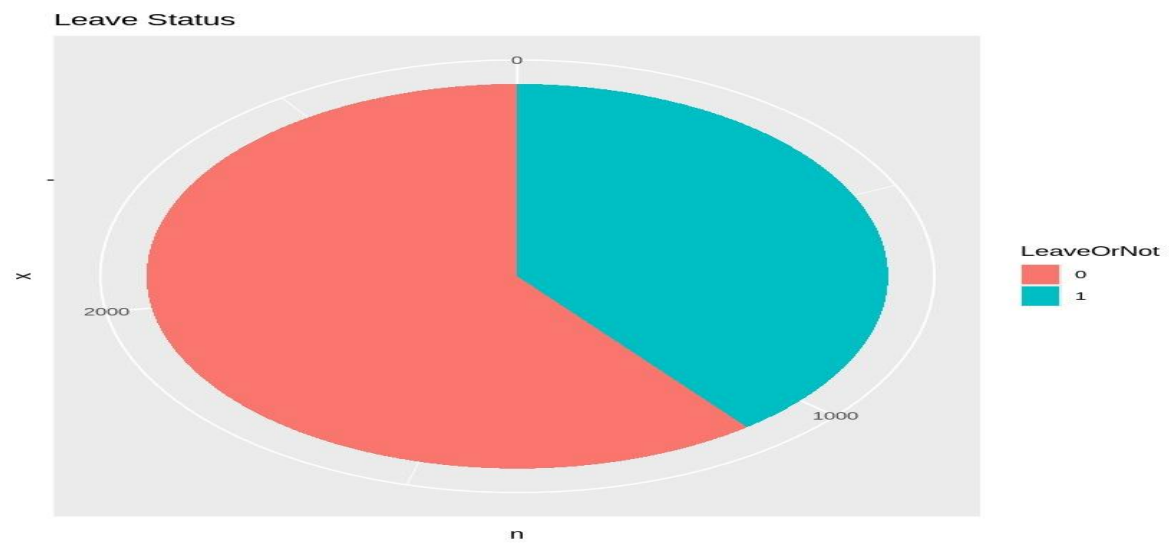
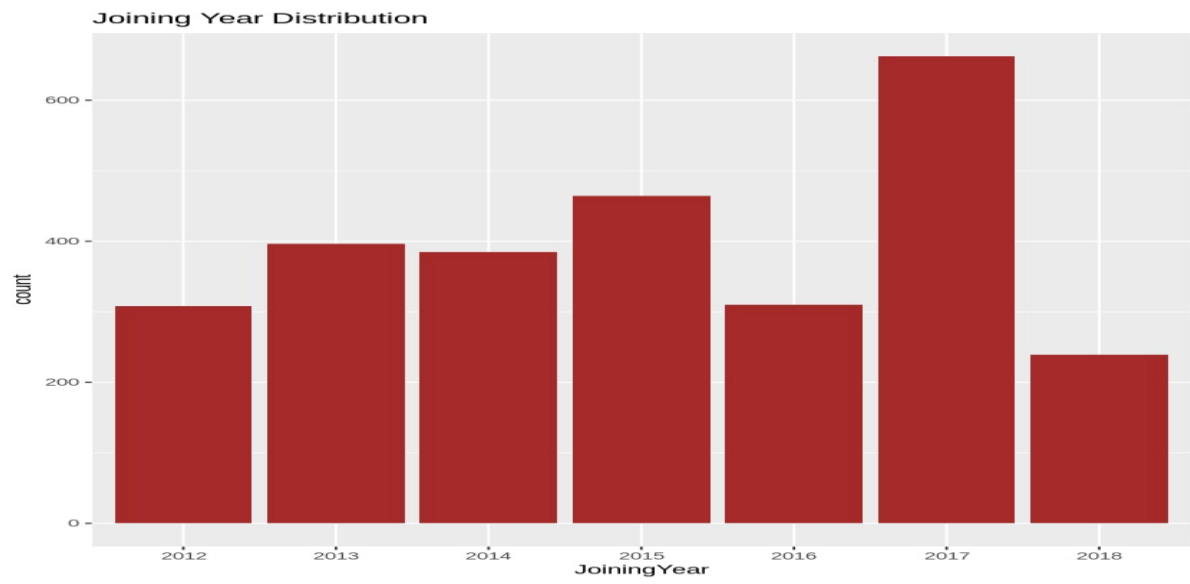
```
summary(employee)
```

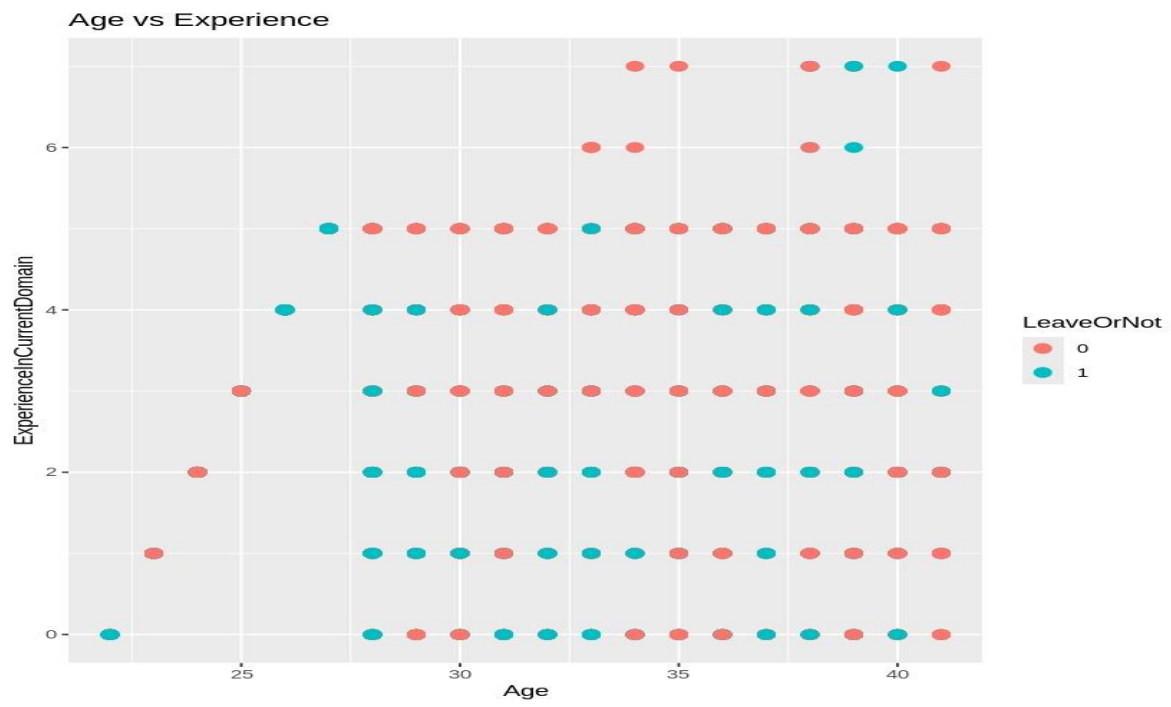
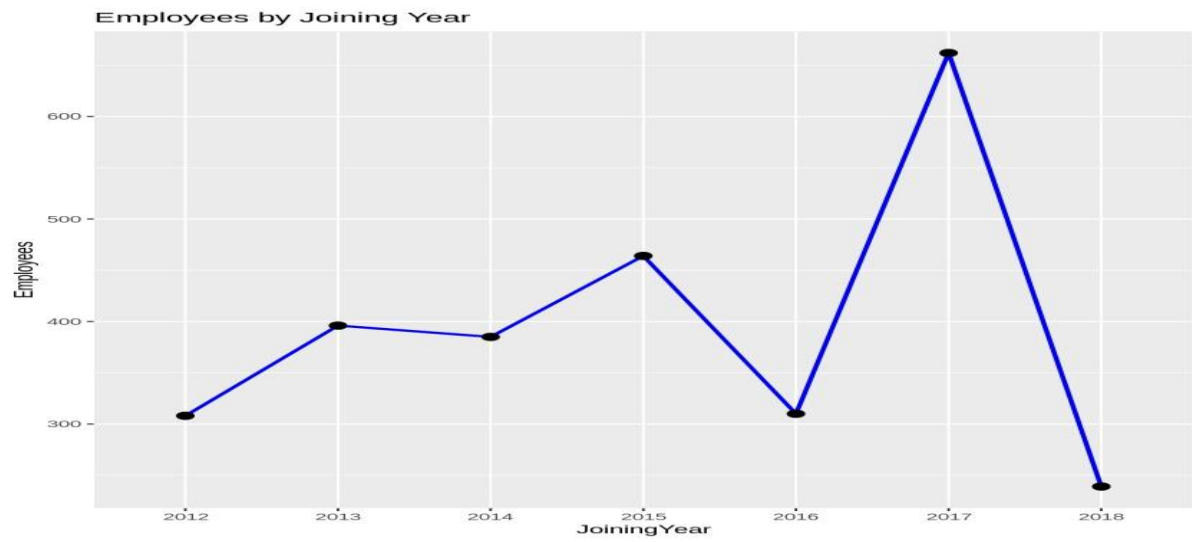
outcome:

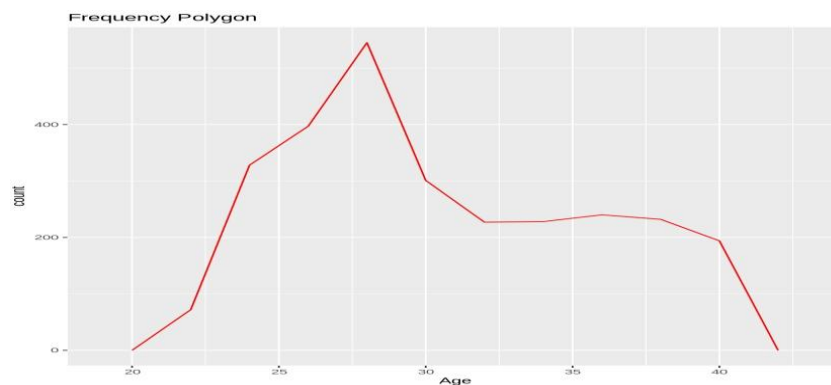
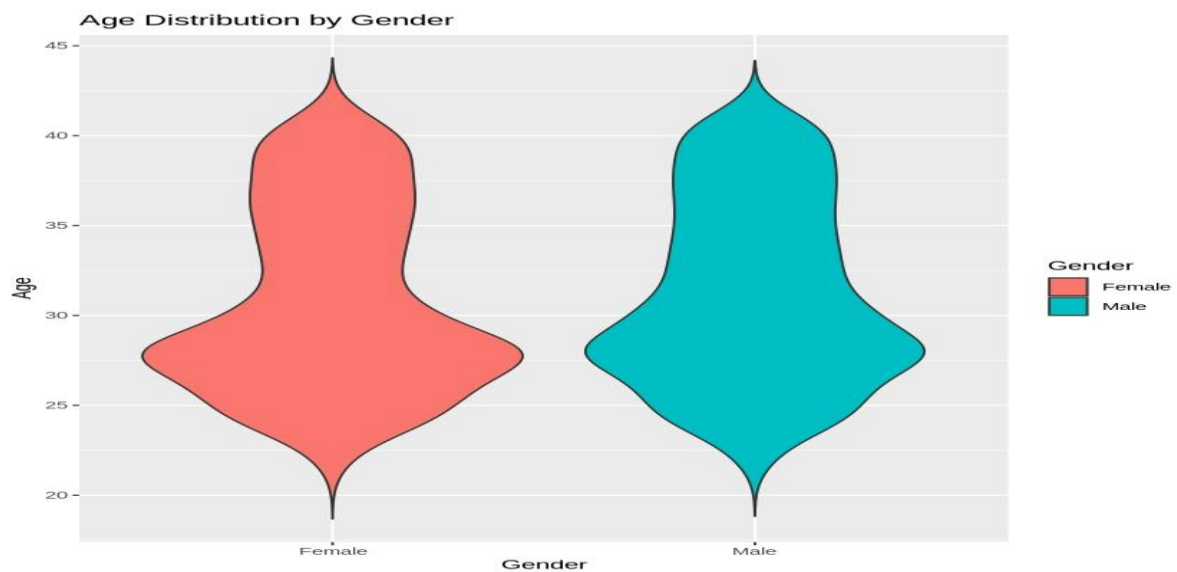
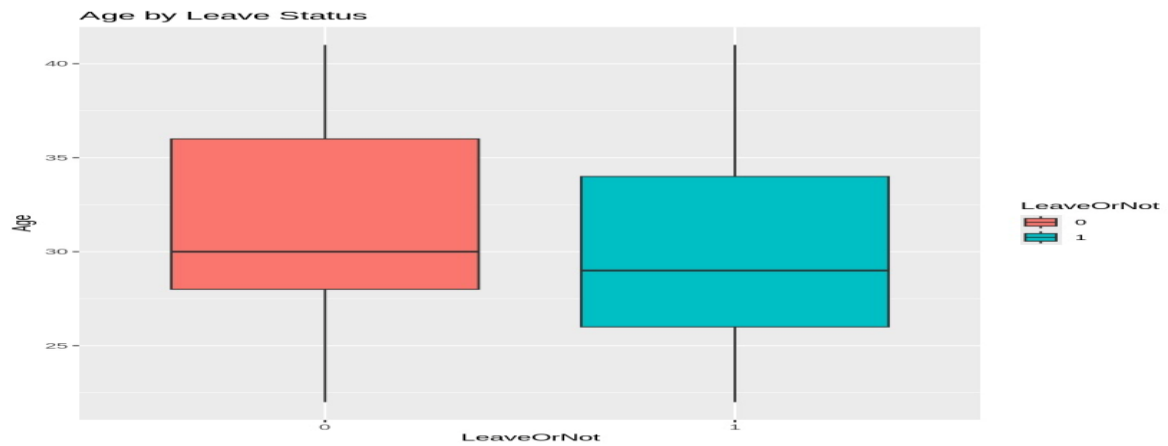












Result

The Employee dataset was successfully imported and analyzed using R. Data handling operations such as checking the dataset structure, summary statistics, and missing values were completed successfully. Various visualizations including Histogram, Bar Chart, Pie Chart, Line Chart, Area Chart, Scatter Plot, Box Plot, Violin Plot, Density Plot, Frequency Polygon, Dot Plot, Q-Q Plot, Stacked Bar Chart, Grouped Bar Chart, Correlation Matrix, and Pair Plot were generated successfully. The visualizations helped identify the distribution of

employee age, education level, gender, city, payment tier, experience, and leave status, providing meaningful insights into the employee data.